

# Discrete Mathematics

## Previous Year Questions — CSE 1st Year 1st Semester

Sylhet Engineering College, Sylhet

Recreated from scanned papers (CSE-15, CSE-16, CSE-17, CSE-18 batches)

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### 1 Topic map

Which topics were asked in which paper. **F** = Semester Final, **TT** = Term Test; the number is the batch (15–18).

| Topic                                                                   | 15 F | 15 TT1 | 15 TT2 | 16 F | 16 TT2 | 17 TT1 | 17 TT2 | 17 F | 18 TT1 | 18 TT2 | 18 F |
|-------------------------------------------------------------------------|------|--------|--------|------|--------|--------|--------|------|--------|--------|------|
| Propositional logic, tautology, equivalences                            |      |        |        | •    |        |        |        | •    |        |        | •    |
| Rules of inference                                                      |      |        |        |      |        |        |        |      | •      |        | •    |
| Sets, Venn diagrams, counting                                           | •    | •      |        |      |        | •      |        | •    |        | •      | •    |
| Functions, floor/ceiling, composition                                   | •    | •      |        | •    | •      |        |        | •    |        | •      | •    |
| Ackermann, recursive definitions                                        | •    |        |        | •    |        | •      |        | •    |        | •      | •    |
| Relations (properties, matrices, digraphs)                              |      |        |        | •    | •      |        |        |      | •      |        | •    |
| Induction, recurrence relations                                         |      |        |        | •    |        |        |        |      |        | •      | •    |
| Graph basics (matrices, isomorphism, planar, connected, colouring)      | •    |        |        | •    | •      |        |        | •    |        |        | •    |
| Graph traversal (BFS, DFS, articulation point)                          | •    |        |        | •    |        |        |        | •    |        |        |      |
| Shortest paths (Dijkstra, Floyd–Warshall)                               | •    |        |        | •    | •      |        | •      | •    |        |        |      |
| Spanning trees (Prim, Kruskal)                                          | •    |        | •      | •    |        |        | •      | •    |        |        | •    |
| Trees (binary, BST, traversals, Huffman, 2-trees)                       | •    |        | •      | •    |        |        | •      | •    |        |        |      |
| Boolean algebra, K-map, logic circuits                                  |      |        |        |      |        |        |        | •    |        |        |      |
| Algorithms & data structures (stack, queue, sorting, searching, Euclid) |      |        | •      |      |        |        |        | •    |        |        |      |
| Permutations                                                            |      |        |        |      |        |        |        | •    |        |        |      |

### Papers found in the scans

| Batch  | Papers                                       | Scan file                             |
|--------|----------------------------------------------|---------------------------------------|
| CSE-15 | Final 2022, Term Test 1, Term Test 2         | CSE-15_batch_all_question_...         |
| CSE-16 | Final 2023, Term Test 2 (no TT1 in the scan) | [1-1] Semester Final TT 16th batch    |
| CSE-17 | Term Test 1, Term Test 2, Final              | CSE-17_(semester final and term test) |
| CSE-18 | Term Test 1, Term Test 2, Final 2024         | CSE-18 [1-1] TT & Semester Questions  |
| CSE-18 | Final 2024 (same paper as above)             | 2024-25 CSE semester final 1-1        |

## 2 CSE-15 batch

### 2.1 Final Examination, 2022

Source: CSE-15 batch file, Discrete Mathematics final (2 pages).

**Sylhet Engineering College, Sylhet**  
(Shahjalal University of Science & Technology)  
Department of Computer Science & Engineering

**Final Examination, 2022**

**Course No:** CSE 103

**Time:** 02 (Two) hours

1<sup>st</sup> Year 1<sup>st</sup> Semester

**Course Title:** Discrete Mathematics

**Full Marks:** 40

N.B.: (i) Answer any two question from each PART (ii) Use separate answer scripts for each PART (iii) Marks allotted are indicated in the margin (iv) Special Instruction (if any): N/A

#### **PART-A**

(Answer any two question)

- 1) a) Which of these sets are equal:  $\{x, y, z\}$ ,  $\{z, y, z, x\}$ ,  $\{y, x, y, z\}$ ,  $\{y, z, x, y\}$ ? [2]  
 b) List the elements of each set where  $\mathbb{N} = \{1, 2, 3, \dots\}$ . [1 × 3]  
 (i)  $A = \{x \in \mathbb{N} \mid 3 < x < 9\}$   
 (ii)  $B = \{x \in \mathbb{N} \mid x \text{ is even, } x < 11\}$   
 (iii)  $C = \{x \in \mathbb{N} \mid 4 + x = 3\}$   
 c) Determine which of the following sets are finite: [2]  
 (i) Lines parallel to the  $x$  axis.  
 (ii) Letters in the English alphabet.  
 (iii) Integers which are multiples of 5.  
 (iv) Animals living on the earth.  
 d) Each student in Liberal Arts at some college has a mathematics requirement  $A$  and a science requirement  $B$ . A poll of 140 sophomore students shows that: 60 completed  $A$ , 45 completed  $B$ , 20 completed both  $A$  and  $B$ . Use a Venn diagram to find the number of students who have completed: (i) At least one of  $A$  and  $B$ ; (ii) exactly one of  $A$  or  $B$ ; (iii) neither  $A$  nor  $B$ . [1 × 3]
- 2) (a) Adjacency matrix of the directed graph  $G$  with vertices  $X, Y, Z, W$  is given below: [6]

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

- i) Draw the directed graph  $G$ . ii) Find the powers  $A^2$ .
- (b) Plot the following functions: (i)  $\left\lceil \frac{x}{2} \right\rceil$  (ii)  $\left\lfloor \frac{x}{2} \right\rfloor$  [4]
- 3) a) Let the functions  $f: A \rightarrow B$ ,  $g: B \rightarrow C$ ,  $h: C \rightarrow D$  be defined by Fig.-1. Determine if each function is: (i) onto, (ii) one-to-one, (iii) invertible. [1 × 3]

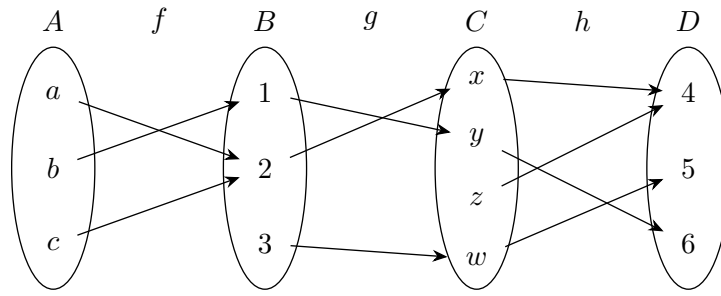


Figure-1

- b) Find the cardinal number of each set: [2]  
 (i)  $A = \{a, b, c, \dots, y, z\}$ , (iii)  $C = \{10, 20, 30, 40, \dots\}$ ,  
 (ii)  $B = \{x \mid x \in \mathbb{N}, x^2 = 5\}$ , (iv)  $D = \{6, 7, 8, 9, \dots\}$ .
- c) Find: (a)  $\lceil 7.5 \rceil$ ,  $\lfloor -7.5 \rfloor$ ,  $\lfloor -18 \rfloor$ ; (b)  $\lceil 7.5 \rceil$ ,  $\lceil -7.5 \rceil$ ,  $\lceil -18 \rceil$ . [2]
- d) **Ackermann function:** [3]  
 (a) If  $m = 0$ , then  $A(m, n) = n + 1$ .  
 (b) If  $m \neq 0$  but  $n = 0$ , then  $A(m, n) = A(m - 1, 1)$ .  
 (c) If  $m \neq 0$  and  $n \neq 0$ , then  $A(m, n) = A(m - 1, A(m, n - 1))$ .  
 Use the definition of the Ackermann function to find  $A(2, 2)$ .

**PART-B**

(Answer any two question)

- 4) a) Define the following term with examples. i) sub-graph ii) complete graph iii) Bipartite graph iv) Regular graph. [2]  
 b) Prove that: The sum of the degrees of the regions of a map is equal to twice the number of edges. [2]  
 c) Find all spanning trees of the graph  $G$  shown in figure-2. [2]

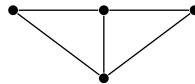


Figure:-2

- d) Find a minimal cost spanning tree  $T$  for the weighted graph  $G$  in Fig.-3 using Kruskal's algorithm. [4]

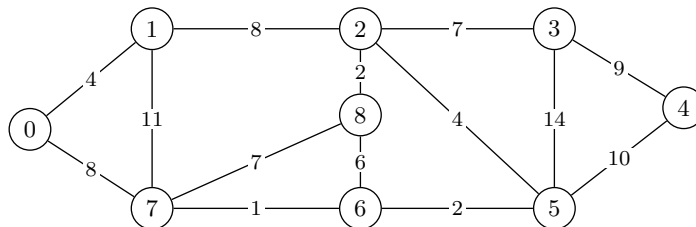


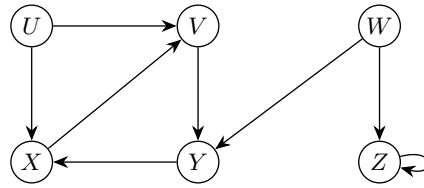
Figure -3

- 5) a) Define vertex coloring and chromatic number. [2]  
 b) Consider the following weighted graph  $G$ . Suppose the nodes are stored in an array DATA as follows: DATA: X, Y, S, T. Find the matrix  $Q$  of shortest paths using Warshall's Algorithm. [4]

[The weighted graph is blacked out in the scan and cannot be recovered.]

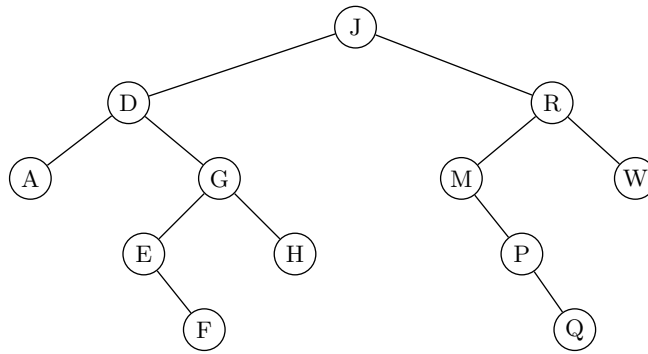
c) Compute the DFS tree for the graph given below.

[4]



6) i) From the following binary search tree insert the nodes C, Z and T.

[6+4]



ii) Find the in-order, pre-order and post-order traversal of the above tree shown in figure.

## 2.2 Term Test-01

Source: CSE-15 batch file (upper half of the term-test page).

### Sylhet Engineering College, Sylhet

Department of Computer Science and Engineering

1<sup>st</sup> year 1<sup>st</sup> Semester Course No: CSE-103 Course Title: Discrete Mathematics

Time: 20 min Total Marks: 15

1. Suppose a list  $A$  contains the 30 students in a mathematics class, and a list  $B$  contains the 35 students in an English class, and suppose there are 20 names on both lists. Find the number of students: (a) only on list  $A$ , (b) only on list  $B$ , (c) on list  $A$  or  $B$  (or both), (d) on exactly one list. [5]
2. Let  $A = \{1, 2, 3\}$ ,  $B = \{a, b, c, d\}$  and  $C = \{c, d\}$ . Find:  $(A \times B) \cap (A \times C)$  and  $(B \cap C)$ . [5]
3. Find: (a)  $\lfloor 7.5 \rfloor$ ,  $\lfloor -7.5 \rfloor$ ,  $\lfloor -18 \rfloor$ ; (b)  $\lceil 7.5 \rceil$ ,  $\lceil -7.5 \rceil$ ,  $\lceil -18 \rceil$ . [5]

## 2.3 Term Test-02

Source: CSE-15 batch file (lower half of the same page order as scanned).

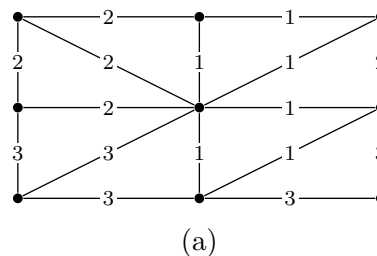
### Sylhet Engineering College, Sylhet

Department of Computer Science and Engineering

1<sup>st</sup> year 1<sup>st</sup> Semester Course No: CSE-103 Course Title: Discrete Mathematics

Time: 20 min Total Marks: 15

1. Find a minimal spanning tree  $T$  for the weighted graph  $G$  in Fig. (a). [7]



2. Consider the algebraic expression  $E = (2x + y)(5a - b)^3$ . Draw the corresponding 2-tree. Use an arrow ( $\uparrow$ ) for exponentiation, an asterisk ( $*$ ) for multiplication, and a slash ( $/$ ) for division. [5]
3. Define Stacks, Queues, and Priority Queues. [3]

### 3 CSE-16 batch

#### 3.1 Final Examination, 2023

Source: CSE-16 batch file (3 pages).

**Sylhet Engineering College, Sylhet**  
(Shahjalal University of Science & Technology)  
Department of Computer Science & Engineering

**Final Examination, 2023**

**Course No:** CSE 143

**Time:** 03 (Three) hours

1<sup>st</sup> Year 1<sup>st</sup> Semester

**Course Title:** Discrete Mathematics

**Full Marks:** 60

N.B.: (i) Answer any three questions from each PART (ii) Use separate answer scripts for each PART (iii) Marks allotted are indicated in the margin (iv) Special Instruction (if any): N/A

#### PART-A

(Answer any three questions)

1. (a) Let  $p$  = “It is raining today” and  $q$  = “CSE first semester exam will take place.” Write down the above propositional statements in English: (a)  $p \rightarrow q$  (b)  $q \leftrightarrow \neg p$ . [2]  
 (b) Determine whether the statement  $(\neg p \wedge \neg q) \leftrightarrow (q \rightarrow r)$  is a tautology or contradiction or contingency. [5]  
 (c) Show that  $(p \rightarrow q) \wedge (p \rightarrow r)$  and  $p \rightarrow (q \wedge r)$  are logically equivalent by developing a series of logical equivalences. [3]
2. (a) Define Relation. Find the relation determined by the figure-1: [1+2]

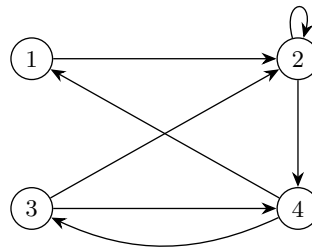


Figure-1

- (b) Given  $A = \{1, 2, 3, 4\}$  and  $B = \{x, y, z\}$ . Let  $R$  be the following relation from  $A$  to  $B$ :  
 $R = \{(1, y), (1, z), (3, y), (4, x), (4, z)\}$  [4]  
 (i) Determine the matrix of the relation.  
 (ii) Draw the arrow diagram of  $R$ .  
 (iii) Find the inverse relation  $R^{-1}$  of  $R$ .  
 (iv) Determine the domain and range of  $R$ .
- (c) Consider the relation  $R = \{(a, a), (a, b), (b, c), (c, c)\}$  on the set  $A = \{a, b, c\}$ . Find: (i) reflexive( $R$ ); (ii) symmetric( $R$ ); (iii) transitive( $R$ ). [3]
3. (a) Find out value of given expressions.

$$\left\lfloor \frac{1}{5} \right\rfloor \quad \lceil 2.5 \rceil \quad \lfloor -2.99 \rfloor \quad \left\lceil -\frac{1}{2} \right\rceil$$

[2]

- (b) If  $f(x) = x^2 + 1$  is a function from  $\mathbb{R}$  to  $\mathbb{R}$ . Is it a one-to-one correspondence? [4]

- (c) Let  $f$  and  $g$  be the functions from the set of integers to the set of integers defined by  $f(x) = 2x + 3$  and  $g(x) = 3x^2 + 2$ . Determine  $f \circ g(3)$  and  $g \circ f(1)$ . [4]

4. (a) Use mathematical induction to show that

$$\sum_{p=1}^n p^3 = \frac{n^2(n+1)^2}{4}$$

for all nonnegative integers  $n$ . [5]

- (b) Suppose that  $f$  is defined recursively by  $f(0) = 0$ ,  $f(1) = 1$ ,  $f(n) = f(n-1) + f(n-2)$ . Find out the value of  $f(7)$ . [5]

### **PART-B**

(Answer any three questions)

5. (a) If on the particular given graph in Fig. 2 if you execute DFS algorithm write the sequence in which nodes are getting traversed: [4]

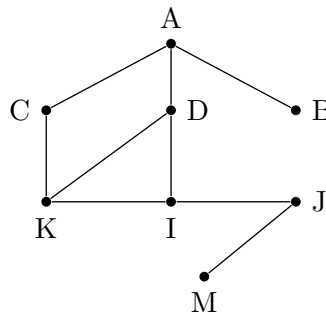


Fig. 2

- (b) Define Articulation Point. Find the articulation point for Fig. 3: [1+3]

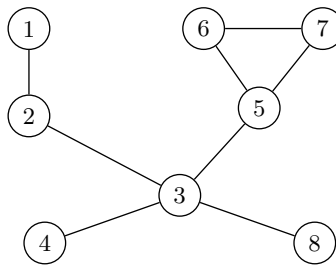


Fig. 3

- (c) Let  $G$  be the directed graph in Fig 4. (a) Find the indegree and outdegree of each vertex of  $G$ . (b) Find the successor list of each vertex of  $G$ . [1+1]

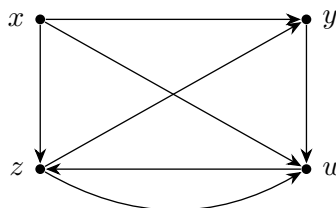
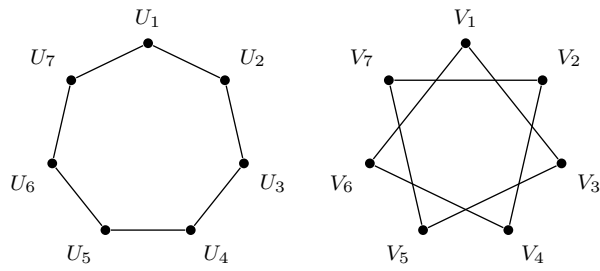
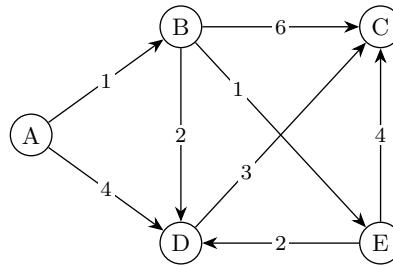


Fig. 4

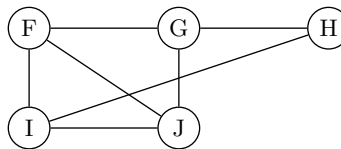
6. (a) Determine whether the given pair of graphs are isomorphic or not? [5]



- (b) In the given graph below, the source node is  $A$  and destination node is  $C$ . Determine the shortest path from  $A$  to  $C$  using Dijkstra algorithm. [5]



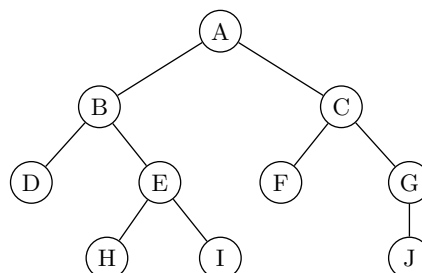
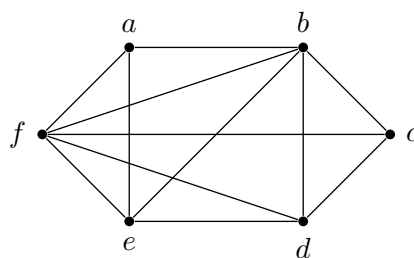
7. (a) Is the given graph a Planar graph? If so, explain why. [2]



- (b) Determine whether the given graphs are connected graph? [4]



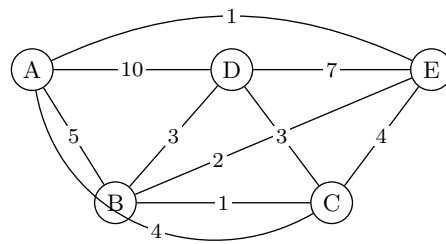
- (c) Using graph coloring theorem, check whether the graph is a bipartite graph or not. [4]



8.

Fig. 1: Binary Tree

- (a) In the above binary tree, what are the ancestors, descendants and sibling node of vertex  $E$ . [3]
- (b) Write down the preorder traversal of the binary tree in figure 1. [3]
- (c) Using Prim's algorithm find a minimum spanning tree of the given graph. [4]



### 3.2 Term Test-2

Source: CSE-16 batch file (last page).

**Sylhet Engineering College, Sylhet**  
Dept. of Computer Science & Engineering

**Term Test -2**

**Course Title:** Discrete Mathematics (CSE 103)

**Full Marks:** 20

**Time:** 40 mins

1. Determine whether the relation represented by the matrix below is reflexive, symmetric or antisymmetric?

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

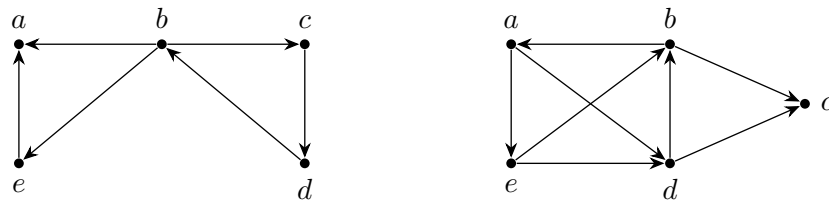
[3]

2. Is every function a relation and vice versa?

[3]

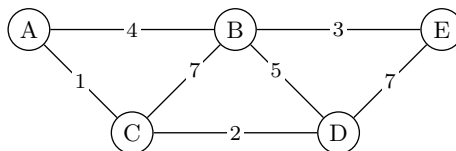
3. Determine whether the given graphs are strongly connected or weakly connected?

[4]



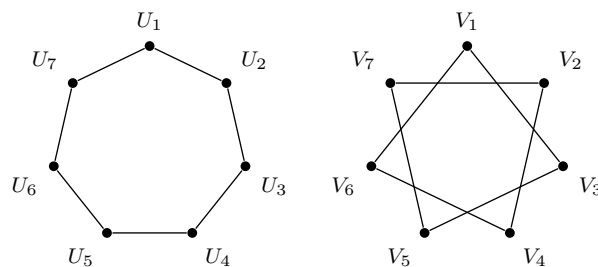
4. In the given graph below, the source node is A and destination node is E. Determine the shortest path from A to E using Dijkstra algorithm.

[5]



5. Determine whether the given pair of graphs are isomorphic or not?

[5]



## 4 CSE-17 batch

### 4.1 Term Test-1

Source: CSE-17 file, first page.

**Sylhet Engineering College, Sylhet**  
Dept. of Computer Science & Engineering

#### Term Test – 1

**Course No:** CSE - (DM) [number not printed]

**Full Marks:** 20

**Time:** 30 mins

1. Define a set and list the different types of sets. Give an example for each type. [2+4]
2. Let  $A = \{1, 2, 3, 4, 5\}$  and  $B = \{3, 4, 5, 6, 7\}$ . Perform the following operations: [8]
  - i)  $A \cup B$    ii)  $A \cap B$    iii)  $A - B$    iv)  $B - A$
3. Write the conditions of the Ackerman function and what will be the ans of  $A(1, 3)$ ? [3+5]

### 4.2 Term Test-2

Source: CSE-17 file, second page.

**Sylhet Engineering College, Sylhet**  
Dept. of Computer Science & Engineering

#### Term Test – 2

**Course No:** CSE - (DM) [number not printed]

**Full Marks:** 20

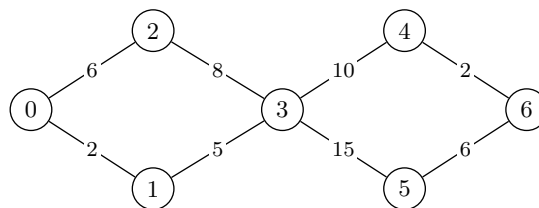
**Time:** 30 mins

1. Apply the Huffman Algorithm to find a 2-tree and also find the Huffman code: [6]

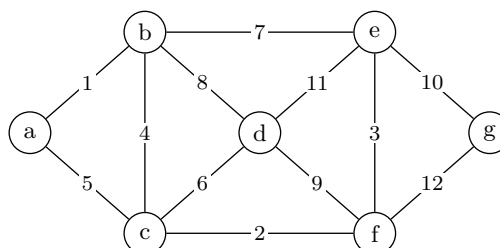
| Node   | A  | B  | C | D  | E  | F  | G |
|--------|----|----|---|----|----|----|---|
| Weight | 10 | 30 | 5 | 15 | 20 | 15 | 5 |

2. [7+7]

- a) Use Dijkstra's algorithm to find the shortest path from zero to any node.



- b) Use the Prim's algorithm to find the minimum cost spanning tree.



### 4.3 Final Examination

Source: CSE-17 file (2 pages). Year not printed.

**Sylhet Engineering College, Sylhet**  
(Shahjalal University of Science & Technology)  
Department of Computer Science & Engineering

#### Final Examination

**Course No:** CSE 0541 1143

**Time:** 03 (Three) hours

1<sup>st</sup> Year 1<sup>st</sup> Semester

**Course Title:** Discrete Mathematics

**Full Marks:** 60

N.B.: (i) Answer all the questions from each PART (ii) Use separate answer scripts for each PART (iii) Marks allotted are indicated in the margin (iv) Special Instruction (if any): N/A

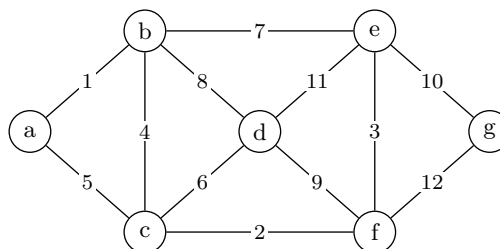
#### Part A

[Answer three questions]

1. a) Define a set and list the different types of sets. Give an example for each type. [02]  
 b) Write the conditions of the Ackerman function and what will be the ans of  $A(1, 3)$ ? [04]  
 c) Out of 100 students, 50 like English, 35 like Science, 29 like Math. 6 like Math and English, 8 like English and Science, 6 like Science and Math. Only 5 like all three subjects. How many students don't like any of these subjects? (Use Venn diagram) [04]
2. a) What is a binary tree give an example. [02]  
 b) Pre-order: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, In-order: 3, 2, 5, 4, 1, 7, 6, 9, 10, 8, 11 using this draw a binary tree. [03]  
 c) Apply the Huffman algorithm to construct the binary Huffman tree and determine the corresponding Huffman codes for each node. [05]

| Node   | A  | B  | C | D  | E  | F  | G |
|--------|----|----|---|----|----|----|---|
| Weight | 10 | 30 | 5 | 15 | 20 | 15 | 5 |

3. a) Define the terms: Euler path, Hamiltonian path, and give an example of where each might be applied in computing. [02]  
 b) What is minimum cost spanning tree? Give an example of subgraph. [02]  
 c) Use Prim's algorithm to find the minimum cost spanning tree. [06]



**OR**

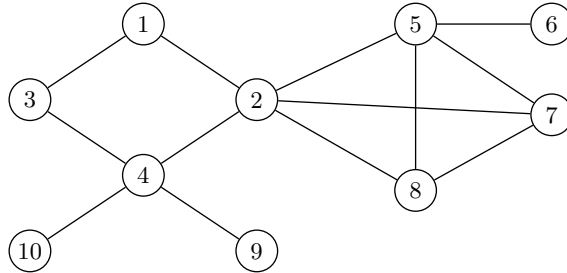
- a) Construct the truth table for  $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$  ( $p \blacksquare q$ ). Is this a tautology?<sup>1</sup> [03]  
 b) How many ways can you arrange the letters in the word *COMPUTE*? [03]  
 c) Write an algorithm to compute the GCD of two positive integers using the Euclidean algorithm. [04]

<sup>1</sup>One connective in the scan is smudged and could not be read; it is shown as  $\blacksquare$ .

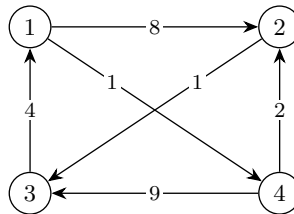
**Part B**

[Answer three questions]

4. a) A function  $f: \mathbb{N} \rightarrow \mathbb{N}$  is defined as  $f(x) = 2x + 1$ . Determine if  $f$  is one-to-one and onto. Justify your answer.<sup>2</sup> [03]  
 b) Use BFS to travel this graph. [07]



5. a) What do you mean by a graph? [01]  
 b) Distinguish between Floyd-Warshall and Dijkstra algorithm. [02]  
 c) Use Floyd-Warshall algorithm to find the shortest path. [07]



6. a) Express the Boolean function  $F = A'B + AB'$  in a minimized form. What logic gate does it represent? [03]  
 b) Design a logic circuit for the function  $F = (A + B)(A' + C)$ . Draw the circuit diagram. [03]  
 c) Explain Merge sort. [04]

**OR**

6. a) Explain binary search. [02]  
 b) Using 2's complement solve the problem  $(-6) + (-10)$ . [04]  
 c) Solve the equation  $xy' + xyz + x'y'z' + x'yz't'$  with K-map. [04]

<sup>2</sup>The sign between  $f(x)$  and  $2x + 1$  is faint in the scan; read as “=”.

## 5 CSE-18 batch

### 5.1 Term Test-01

Source: CSE-18 file, page 14.

#### Sylhet Engineering College, Sylhet

# Term Test – 01    # CSE-1143 (Discrete Mathematics)    # Time: 40 Minutes    # Full Marks: 20

1. Show that the premises “It is not sunny this afternoon and it is colder than yesterday,” “We will go swimming only if it is sunny,” “If we do not go swimming, then we will take a canoe trip,” and “If we take a canoe trip, then we will be home by sunset” lead to the conclusion “We will be home by sunset.” [6]
2. Determine which rule of inference is used in the following arguments:  
If it rains today, then we will not have a barbecue today. If we do not have a barbecue today, then we will have a barbecue tomorrow.  
Therefore, if it rains today, then we will have a barbecue tomorrow. [3]
3. Consider the following five relations on the set  $A = \{1, 2, 3, 4\}$ :

$$R = \{(1, 1), (1, 2), (2, 3), (1, 3), (4, 4)\}$$

$$S = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$$

$$T = \{(1, 3), (2, 1)\}$$

$$U = A \times A, \text{ the universal relation}$$

$$V = \emptyset, \text{ the empty relation}$$

Determine with reason whether or not each of the above relations on  $A$  is i) Reflexive ii) Symmetric iii) Transitive [6]

4. Let  $A = \{1, 2, 3, 4, 5, 6\}$  and let  $R$  be the relation on the set  $A$ ;  $R = \{(x, y) \mid x \text{ divides } y\}$ 
  - i) Write  $R$  as a set of ordered pairs. [2]
  - ii) Draw its directed graph. [3]

### 5.2 Term Test-02

Source: CSE-18 file, page 15.

#### Sylhet Engineering College, Sylhet

# Term Test – 02    # CSE-1143 (Discrete Mathematics)    # Time: 40 Minutes    # Full Marks: 20

1. Find  $f(4)$  if  $f$  is defined recursively by  $f(0) = -1$ ,  $f(1) = 2$  and  $f(n+1) = f(n)^2 f(n-1)$ . [3]
2. Find the solution to the recurrence relation:  $a_n = 2a_{n-1} + 5a_{n-2} - 6a_{n-3}$  with the initial conditions  $a_0 = 7$ ,  $a_1 = -4$  and  $a_2 = 8$ . [7]
3. In a class of 80 students, 50 students know English, 55 students know French and 46 students know German language. 37 students know English and French, 28 students know French and German, 7 students know none of the languages. How many students know all the three languages? [5.5]
4. Define the following function with example: i. Bijective    ii. Ceiling    iii. Floor [4.5]

### 5.3 Final Examination, 2024

Source: CSE-18 file (pp. 16–17) and 2024–25 CSE semester final 1-1.pdf (pp. 3–4); the two scans are the same paper.

**Sylhet Engineering College, Sylhet**  
(Shahjalal University of Science & Technology)  
Department of Computer Science & Engineering

**Final Examination, 2024**

**Course No:** CSE 0541 1143

**Time:** 03 (Three) hours

1<sup>st</sup> Year 1<sup>st</sup> Semester

**Course Title:** Discrete Mathematics

**Full Marks:** 60

N.B.: (i) Answer all questions from each PART (ii) Use separate answer scripts for each PART (iii) Marks allotted are indicated in the margin (iv) Special Instruction (if any): N/A

#### PART-A

1. (a) Define Tautology. Show that  $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$  is a tautology. [03]  
 (b) Show that  $\neg(p \vee (\neg p \wedge q))$  and  $\neg p \wedge \neg q$  are logically equivalent by developing a series of logical equivalences. [05]  
 (c) Determine which rule of inference is used in the following arguments: [02]  
 If it rains today, then we will not have a barbecue today. If we do not have a barbecue today, then we will have a barbecue tomorrow.  
 Therefore, if it rains today, then we will have a barbecue tomorrow.
2. (a) Define Ackermann function. From the definition of that function find  $A(1, 3)$ . [Hints.  $A(m, n)$ ] [05]  
 (b) Prove the following: i)  $A \cap (A \cap B) = A$  ii)  $A \cap (A \cup B) = A^3$  [05]
3. (a) Define reflexive, symmetric and transitive relation. [03]  
 (b) Consider the following five relations on the set  $A = \{1, 2, 3, 4\}$ : [05]

$$R = \{(1, 1), (1, 2), (2, 3), (1, 3), (4, 4)\}$$

$$S = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$$

$$T = \{(1, 3), (2, 1)\}$$

$$U = A \times A, \text{ the universal relation}$$

$$V = \emptyset, \text{ the empty relation}$$

Determine whether or not each of the above relations on  $A$  is i) Reflexive ii) Symmetric iii) Transitive.

Define Equivalence relation with example. [02]

#### OR

3. (a) Let  $A = \{1, 2, 3, 4, 5, 6\}$  and let  $R$  be the relation on the set  $A$ ;  $R = \{(x, y) \mid x \text{ divides } y\}$   
 i) Write  $R$  as a set of ordered pairs. [02]  
 ii) Draw its directed graph. [02]  
 (b) Draw the graph  $G$  corresponding to the adjacency matrix: [03]

$$A = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}$$

<sup>3</sup>Printed this way in the paper; part (i) looks like a typo for  $A \cup (A \cap B) = A$ .

- (c) What is isomorphism? Draw two graphs that are isomorphic and describe briefly. [03]

**PART-B**

4. (a) Find the solution to the recurrence relation:  $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$  with the initial conditions  $a_0 = 2$ ,  $a_1 = 5$  and  $a_2 = 15$ . [05]

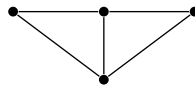
- (b) Define with figure: i. Regular graph ii. Bipartite graph iii. Complete graph [05]

5. (a)  $f, g, h$  are functions on  $X = \{1, 2, 3\}$  as

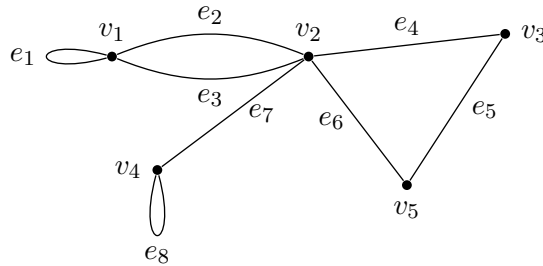
$$f = \{(1, 2), (2, 3), (3, 1)\}, \quad g = \{(1, 2), (2, 1), (3, 3)\}, \quad h = \{(1, 1), (2, 2), (3, 1)\}$$

Compute i)  $f \circ g \circ h$  ii)  $f \circ h \circ g$  [04]

- (b) Find all spanning trees of the following graph: [04]



- (c) Represent the graph with an incidence matrix: [02]



6. (a) Consider Universal set  $U$  and two sets  $A$  and  $B$ . Shade the following sets using Venn diagram: i.  $A \cap B^C$  ii.  $(B/A)^C$  [04]

- (b) Define the following function with example: i. Bijective ii. Ceiling iii. Floor [06]

**OR**

6. (a) Find the value of  $A(1, 3)$  using Ackermann function. [04]

- (b) In a class of 80 students, 50 students know English, 55 students know French and 46 students know German language. 37 students know English and French, 28 students know French and German, 21 students know English and German, 7 students know none of the languages. How many students know all the three languages? [04]

- (c) Find  $f(4)$  if  $f$  is defined recursively by  $f(0) = -1$ ,  $f(1) = 2$  and  $f(n+1) = f(n)^2 f(n-1)$ . [02]

## 6 Notes on the scans

- Graphs and diagrams were redrawn from the scans, so vertex positions differ from the originals. Edge sets and weights follow the scans.
- CSE-15 Final Q5(b): the graph is blacked out in the scan and is left as a placeholder.
- CSE-17 Final, Part A OR (a): one connective in the truth-table expression is smudged.
- CSE-17 papers: the course number and the exam year are not printed on the term tests; the final gives no year.
- CSE-16: only Term Test 2 appears in the scanned file.
- 2024 Final Q2(b)(i) is reproduced exactly as printed.
- Some 2–3 mark values in CSE-15 Q4 and Q5 are read from the margin and may be off by a mark.